

Asynchronous Agreement

- ❑ Problem 1: A1 and A2 have to agree who goes down the hill first

★ Solution:

- both send message with number of team members,
- wait until receive the message of the other group;
- The group that has more members goes down the hill first

As algorithm:

- For team A_i , $i=1,2$
 - ▲ SA_i = size of team A_i
 - ▲ Send(SA_i) to other
 - ▲ Wait for message (SA_j) from other
 - ▲ If $SA_i > SA_j$: First := A_i
 - ▲ Else if $SA_j > SA_i$: First := A_j
 - ▲ ELSE First: = A_1

Asynchronous Agreement

- ❑ Problem: A1 and A2 have to agree **when** to go down the hill; they have to do this nearly at the same time (difference at most X time-units)

Assume A1 has is the first

- ❑ Asynchronous:
 - ☆ A1 makes suggestion “at 2pm” or “in 20 minutes”.
 - ☆ But no guarantee that message arrives before 2 pm or within 20 minutes.
- ❑ Synchronous model:
 - ☆ A1 makes suggestion “go now”, waits for *min* time units and goes down
 - ☆ After receipt of message, A2 goes down the hill
 - ☆ A2 goes down the hill at most (*max* - *min*) time units after A1
 - ☆ If $(\text{max} - \text{min}) \leq X$, this works!

Algorithm

- ❑ Assume A1 should go down the hill first
- ❑ A1:
 - ★ Send (start!) to A2
 - ★ Wait Min time units
 - ★ Go down the hill
- ❑ A2:
 - ★ Wait (start!) from A1
 - ★ Go down the hill